



Department of Electronics and Communication Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch : III Semester B.E. CSE

Course Code & Title : CS8351 Digital Principles and System Design

Name of the Faculty member : Mr.A.Rameshbabu

Innovative Practice Description

- **Unit / Topic:** Unit IV / Races and Hazards
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 19
- **Activity Chosen:** Flipped Class room
- **Justification:**
 - A flipped classroom is a type of blended learning where students are introduced to content at home and practice working through it at College.
- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**
 - This is the reverse of the more common practice of introducing new content at college, then assigning homework and projects to complete by the students independently at home.
 - In this blended learning approach, face-to-face interaction is mixed with independent study—usually via technology. In a common Flipped Classroom scenario, students might watch pre-recorded videos at home, then come to school to do the homework armed with questions and at least some background knowledge.
 - The concept behind the flipped classroom is to rethink when students have access to the resources they need most. If the problem is that students need help doing the work rather than being introduced to the new thinking behind the work, then the solution the flipped classroom takes is to reverse that pattern.
- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 4	PO 9	PO 10	PO 12	PSO 1
CO3	2	2	3	1	1	1	1	1

(1 – Low

2 – Moderate

3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 9	PO 10	PSO 1
Justification for correlation	Students will function effectively as an individual, and as a member or leader in Open Flipped Classroom activity to find hazard in combinational & sequential circuits, So Course outcome is mapped at level 1	In Flipped Classroom activity students will make effective presentations on asynchronous sequential circuits concepts, hence Course outcome is mapped at level 1	Races and Hazards topic used to solve complex computation task of asynchronous sequential circuits, So Course outcome is mapped at level 1

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students eagerly participated and enjoyed this activity.

❖ **Benefit of the practice:**

- Outcome attainment would have increased due to innovative practice over conventional practice.
- Students easily understand the concepts Race Condition in Asynchronous Sequential Circuits.
- Students can answer easily about Hazard concepts and they can design Hazard Free Circuits.

❖ **Challenges faced in implementation:**

- Faced issues while make the students to understand about Flipped Class room Activity rules.

References:

- ❖ <https://www.electrically4u.com/hazards-in-digital-circuits/>
- ❖ <https://www.geeksforgeeks.org/static-hazards-in-digital-logic/>
- ❖ <https://www.youtube.com/watch?v=YgCcfTBXMwo>
- ❖ <https://www.allaboutcircuits.com/technical-articles/hazards-in-combinational-logic/>



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Innovative Practice Description

- **Unit / Topic:** Unit III / Realization of one Flip Flop to other Flip Flops, Master Slave Flip Flop
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO 12
- **Activity Chosen:** Collaborative Learning - Think-Pair-Share
- **Justification:**
 - Collaborative learning is the educational approach of using groups to enhance learning through working together. Groups of two or more learners work together to solve problems, complete tasks, or learn new concepts.
- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**
 - Collaborative learning strategies are explicit approaches or procedures to guide the process of collaborative learning.
 - Collaborative learning occurs when dyads or small groups have been engineered to share responsibility, authority, and learning outcomes.
 - Collaborative learning strategies provide the frameworks and step-by-step processes to facilitate interdependence among group members, active participation, interactive dialogue, and co-creation of academic products, all of which are hallmarks of collaborative learning.
 - In essence, these strategies are the architecture or infrastructure to facilitate construction of knowledge among students.

• **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 10	PO 12	PSO 1	PSO 2
CO3	2	2	3	2	1	1	1	1	1

(1 – Low

2 – Moderate

3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 3	PO 10	PSO 1	PSO 2
Justification for correlation	Students will determine the design objectives and functional requirements of the synchronous sequential circuits, So outcome is mapped at level 3	In Collaborative Learning activity students will make effective presentations on flip flops concepts, hence outcome is mapped at level 1	Students will be able to solve complex computation task with fundamental design procedures of Clocked Sequential Circuits soft skills, So outcome is mapped at level 1	Analysis and Design Procedures of Clocked Sequential circuits helps to develop reliable IT solutions with combinational logic soft skills, hence Course outcome is mapped at level 1

• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students eagerly participated and enjoyed this activity.

❖ **Benefit of the practice:**

- Outcome attainment would have increased due to innovative practice over conventional practice.
- Students easily understand the Realization of Flip Flops concepts.
- Students can answer easily about Master Slave Flip Flop concepts.

❖ **Challenges faced in implementation:**

- Faced issues while make the students to understand about Collaborative Learning Activity rules.

References:

- ❖ <https://www.valamis.com/hub/collaborative-learning>
- ❖ <https://www.youtube.com/watch?v=7gWS4mxM1Qc>
- ❖ <https://www.electronics-tutorials.ws/sequential/conversion-of-flip-flops.html>



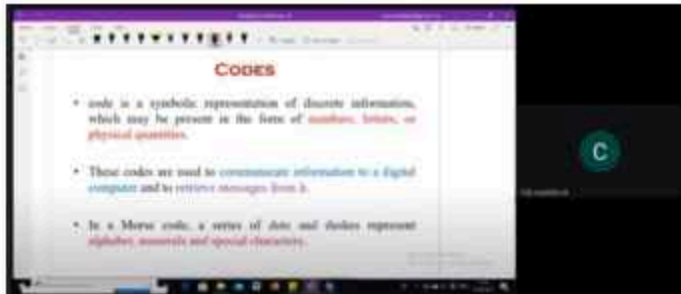
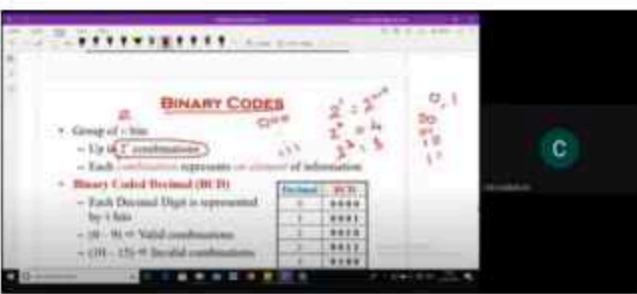
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INNOVATIVE TEACHING METHODS

Degree, Semester & Branch : III Semester B.E. CSE

Course Code & Title : CS8351 Digital Principles and System Design

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Sl. No.	Date	Topic(s)	Activity*	Reference																																								
UNIT I - BOOLEAN ALGEBRA AND LOGIC GATES																																												
1.	23.08.21	Number Systems	Class Poll	https://www.polleverywhere.com/classroom-response-system																																								
Exercise – Convert ... <table><thead><tr><th>Decimal</th><th>Binary</th><th>Octal</th><th>Hexa-decimal</th></tr></thead><tbody><tr><td>33</td><td></td><td></td><td></td></tr><tr><td></td><td>1110101</td><td></td><td></td></tr><tr><td></td><td></td><td>703</td><td></td></tr><tr><td></td><td></td><td></td><td>1AF</td></tr></tbody></table>			Decimal	Binary	Octal	Hexa-decimal	33					1110101					703					1AF	Answers <table><thead><tr><th>Decimal</th><th>Binary</th><th>Octal</th><th>Hexa-decimal</th></tr></thead><tbody><tr><td>33</td><td>100001</td><td>41</td><td>21</td></tr><tr><td>117</td><td>1110101</td><td>165</td><td>75</td></tr><tr><td>451</td><td>111000011</td><td>703</td><td>1C3</td></tr><tr><td>431</td><td>110101111</td><td>657</td><td>1AF</td></tr></tbody></table> 😊		Decimal	Binary	Octal	Hexa-decimal	33	100001	41	21	117	1110101	165	75	451	111000011	703	1C3	431	110101111	657	1AF
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431	110101111	657	1AF																																									
2.	25.08.21	Binary Codes	Mind Map	https://www.mindmeister.com/blog/students-guide-to-mind-mapping/																																								
																																												
3.	07.09.21	Logic Gates - NAND and NOR Implementations	One Minute Paper	https://oncourseworkshop.com/self-awareness/one-minute-paper/																																								
																																												



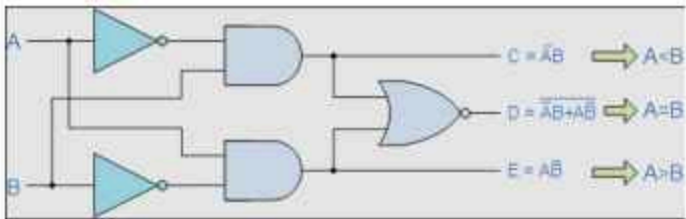
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UNIT II - COMBINATIONAL LOGIC				
1.	23.09.21	Magnitude Comparator	Card Clusters	https://gdhr.wa.gov.au/learning/teaching-strategies/tuning-in / card-clusters
				
2.	25.09.21	Encoders	Head and Talk	https://gdhr.wa.gov.au/learn ing/teachingstrategies/finding -out/head-talk
				
3.	01.10.21	Demultiplexer	One Minute Paper	https://oncourseworkshop.co m/self-awareness/one- minute-paper/
				



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UNIT III - SYNCHRONOUS SEQUENTIAL LOGIC				
1.	09.10.21	Flip Flops (JK,T)	Write-Pair-Share	https://www.readingrockets.org/strategies/think-pair-share
				
2.	25.09.21	Encoders	Head and Talk	https://gdhr.wa.gov.au/learning/teachingstrategies/finding-out/head-talk
				
3.	01.10.21	Demultiplexer	One Minute Paper	https://oncourseworkshop.com/self-awareness/one-minute-paper/
				



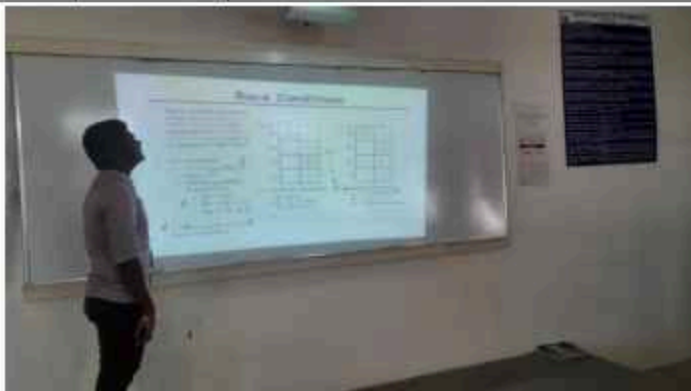
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UNIT IV - ASYNCHRONOUS SEQUENTIAL LOGIC				
1.	09.11.21	Analysis of Asynchronous Sequential Circuits - mealy model	One Minute Challenge	https://gdhr.wa.gov.au/learning/teachingstrategies/tuning-in/one-minute-challenge
 				
2.	16.11.21	Race-free State Assignment – Multiple Rows method	Fishbowl Debate	https://www.facinghistory.org/resource-library/teaching-strategies/fishbowl
 				
3.	18.11.21	Hazards: hazard in sequential circuits	Open Book Quiz	https://www.educationcorner.com/openbook-tests.html
 				



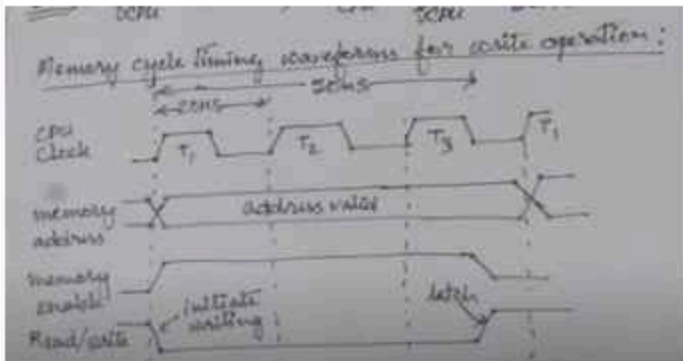
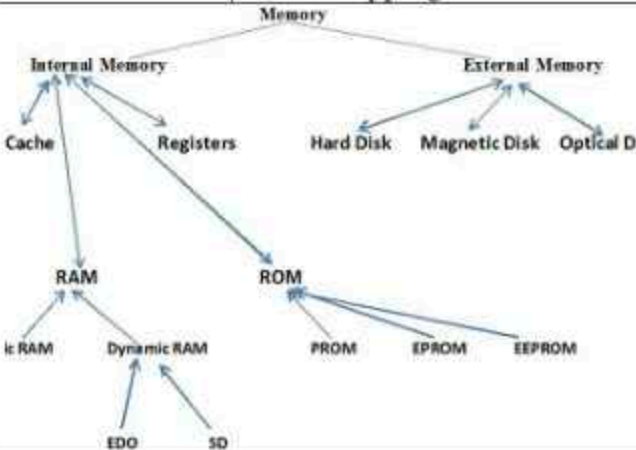
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Sl. No.	Date	Topic(s)	Activity*	Reference
UNIT V - MEMORY AND PROGRAMMABLE LOGIC				
1.	18.11.21	RAM – RAM organization, Write & Read Operation	One Minute Challenge	https://gdhr.wa.gov.au/learning/teachingstrategies/tuning-in/one-minute-challenge
				
2.	24.11.21	ROM types: PROM – EPROM – EEPROM – EAPROM	Mind map	https://www.mindmeister.com/blog/students-guide-to-mind-mapping/
				
3.	24.11.21	Sequential Programmable Devices	Write-Pair-Share	https://www.readingrockets.org/strategies/think-pair-share
		